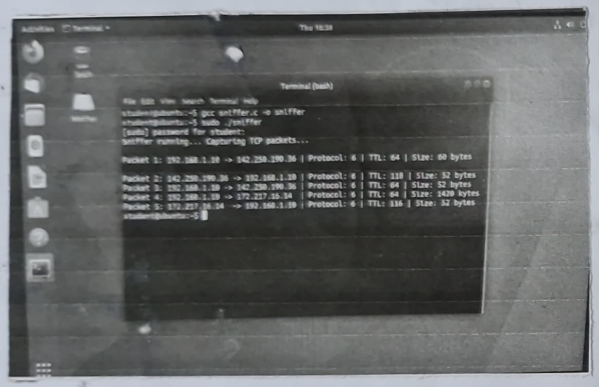


EXP-14 Write a code for RAW sockets. packet sniffing.

- Aim:
  To implemented a network. packet sniffer. in C using RAM sockets to capture, decode, and display network. layer packets

- C-code implementation:

```
#include <stdio.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <netinet/ip.h>

int main()
  int sock = socket (af_net, sock_raw, iproto_tcp);
  if (sock < 0) return 0

unsigned char buf [65536];
struct sockaddr_in saddr, daddr;
printf ("capturing TCP packet");

for (int i=0 ; i<5 ; i++){
  int len = recvfrom (sock, buf, sizeof(buf), 0, NULL, NULL);
  if (len < 0) continue;
  struct iphdr *ip = (struct iphdr *) buf;
  saddr.sin_addr.s_addr = ip->saddr;
  daddr.sin_addr.s_addr = ip->daddr;

  printf ("%s -> %s | Protocol: %d | TTL: %d | Size: %d bytes\n",
  inet_ntoa (saddr.sin_addr), inet_ntoa (daddr.sin_addr),
  ip->protocol, ip->ttl, len);

}
close (sock);
return 0;
```

- Result:
  Hence packet sniffing was implemented in C programming using RAW sockets.